then recalled into the housing via some containment field. The energy is finally treated to a final state to trickle to the battery and complete one hell of a circuit. This looks great on a blueprint but present day batteries are in no way powerful enough to generate plasma, not to mention material science will have to make generation leaps to house this high temperature equipment and create focusing crystals for precise beams. Plasma solves the two problems posed by lasers. It will interact with another plasma based beam like a lightsaber should, and it is possible to contain plasma to form a limited length beam but the technology hasn't quite caught up with our dreams (or rather George Lucas's).

Blasters : What are they anyway?

An inelegant weapon yes, but at close range even the most skilled Jedi appreciate the power of a blaster. But what are these blasters? They are certainly not laser guns. Like we discussed in the context of lightsabers, a laser beam basically consists of photons which travel at the speed of light. The beam from the blasters however moves much much slower than that. For one thing we can "see" those projectiles and the Jedi can even deflect them using their lightsabers. Interestingly, the US Navy has deployed a laser weapon on one of its ships : The USS Ponce. Inspired by Star Wars perhaps?

The typical Star Wars blaster however uses a fictitious energy rich gas in a chamber that is somehow supercharged with energy and converted to a coherent light filled beam or "bolt". While this sounds like a simple construction there are plenty of engineering issues with the current available technology to develop a similar design. Dealing with the heat released on converting gas to energy, actuating the gas and forming a coherent beam and finally, coming up with a compact power source to achieve the above design. Beyond all this let's ask a very important question: is a Blaster a practical weapon? Analysis suggests the speed of the blaster 'bolt' is less than a bullet. So why would you want to build one? It's simple. A Blaster is cooler than a machine gun. Cooler trumps quicker and here in our discussion of Star Wars tech cooler is always better.

Interstellar travel : Hyperspace and it's unbelievable physics

Time and again in the Star Wars universe space ships cover incredibly large distances in relatively short time. By incredibly large we mean distances where light would take years to reach and by relatively short time we mean time scales accessible to mortal humans. How are these trips undertaken? Does physics have any explanation for these seemingly impossible journeys or are they simply a lazy writer's plot device to allow interstellar travel.

According to Einstein's special theory of relativity the speed of light is invariant so that sets a bound on the maximum speed a spaceship can theoretically achieve. The closer the space ship will get to lightspeed, the slower time will get and the more mass it will gain. Eventually the mass of the spaceship will inflate to an absurdly large number that cannot be powered for a journey in any conceivable way, not to mention that on return Han Solo's kids would have aged more than him.

Too inelegant to be a Jedi weapon

So is there some way to cheat Einstein's special theory of relativity and somehow circumvent this speed limit imposed by it. Apparently, we can and the cheat code is hidden in Einstein general theory of relativity.

Now it is well know that Han uses 'hyperspace' for interstellar travel. What hyperspace might be is a hotly debated topic, not only in the science fiction geek community but also on the physics nerd table (Both groups have significant overlap of course). You see, space and time are not separate entities but are intricately linked and form the fabric of the entire universe. We call this fabric as spacetime. Think of spacetime as a two dimensional sheet stretched till the farthest eons of space and time. Now imagine two dots on the sheet, lets call them Tatooine and Alderaan. The dot between those two is the Millennium Falcon. Now imagine if the sheet allows expansion and compression, we could expand the gap between the Falcon and Tatooine and compress the gap between the ship and Alderaan. The points haven't moved anywhere absolutely but still the Falcon is closer to Alderaan! This is essentially what a warp drive is imagined to do, expand spacetime behind and contract spacetime in front of a space vehicle to basically bring the destination to the vehicle enclosed in a sort of spacetime bubble. We've spoken about the Alcubierre drive many times before so we won't bore you.

Another explanation for the casual interstellar travel depicted in Star Wars is the worm hole explanation. Ironically "Interstellar" did a good job of explaining it with the folded paper and holes so we won't go into it here. Why haven't we done it yet? Manipulating spacetime requires gargantuan amounts of energy, a scale we aren't even close to reaching. And though elegant these aren't the techniques the Melamine Falcon probably uses. A possible option is that Han actually accesses one of the many dimensions curled up in the universe (as proposed by String Theory) which allows him take shortcuts that wouldn't be possible in a simple three dimensional world. All this sounds simple in theory but all this is as close to fiction as science can get.

Flight Dynamics and Star Wars : Not exactly a match made in heaven

Star Wars has an awesome collection of space vehicles, fighters and ships. Like in any other science fiction world these craft are powerful machines and capable of executing complex manoeuvres. But do they hold true to the laws of Physics of the known universe? Let's find out.

Let's start of with a refresher course in aerodynamics. All terrestrial aircraft need to generate lift to stay afloat. Hence, they are equipped with wings that essentially push against air and create a pressure difference that creates lift. You know where we're leading up to... Yes wings are redundant in the vacuum of space. In space a vehicle need only use directional thrusters